

LECTURE ACTIVITY 11: LEND ME A HAND – UPDATED – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Learning Objectives:

By the end of this activity, you should be able to

- 1) Approximate and interpret a 95% confidence interval for a population proportion by using the 2SD method.
- 2) Compute and interpret a confidence interval for a population proportion using a theory-based approach, including checking validity conditions.
- 3) Infer how each part of the confidence interval changes when the confidence level or sample size change

Key Terms:

- Confidence Interval
 - Standard Error
 - Significance Level
 - 2SD Method
 - Confidence Level
-

Intro

What proportion of US adults¹ are right-handed? This is the question we're going to explore today. Recall that, as a part of the Welcome Survey I (Dr. Ethan) sent out to all three of my STAT 1110 sections, 69 out of 84 respondents indicated they are right-handed.



Question 1

Identify the population and sample in this survey.

Solution: The population is the set of all US adults, and the sample is the 84 STAT 1110 students that responded to the welcome survey.



Question 2

Is it reasonable to believe that the sample of 84 respondents is representative of the larger population? Explain why or why not.

Solution: Technically, we obtained a convenience sample so we should be a little cautious as we know convenience sampling can be a biased sampling method. However, there is no reason to believe STAT 1110 students differ in handedness from the broader population of US residents, so we are actually probably okay to proceed with inference (albeit, potentially with some caution).

¹We'll define "adult" to mean aged 18 or older

 Question 3

Is the value of $69/84 = 82.14\%$ a statistic or a parameter? What symbol should we use to denote it? Explain.

- This quantity is a: ☐ Parameter ☒ **Statistic**
- The correct symbol is: $\hat{p}$

Here is my explanation:

Solution: The value of 82.14% was obtained from the *sample*, meaning it is a statistic. Furthermore, it is a sample *proportion*, meaning we use the symbol $\hat{p}$ to denote it.

 Question 4

Identify, in words and symbols, the parameter of interest.

Solution: The parameter of interest is the population proportion of all US residents that are right-handed, which we denote by π (or π_R , or similar notation).

 Question 5

What is the variable of interest in this study? Is it categorical or quantitative? If it is categorical, please phrase it as a binary variable and clearly indicate the levels.

Solution: The variable is **whether a given person is right-handed or not**, which is a **categorical** variable with levels “**Yes**” or “**No**.” **Note:** I’d encourage you to NOT think of the variable as “handedness” as some people are ambidextrous! So, “handedness” as a variable really has *three* levels and is therefore not binary.

 Question 6

Is it reasonable to conclude that exactly 82.14% of all US adults are right-handed? Why or why not?

Solution: **No**, it is not reasonable to conclude this. The sample proportion of 82.14% is only an *estimate* of the population proportion; different random samples from the population will likely give slightly different results.

Though we might expect the true value of the parameter to be close to 82.14%, we realize there may be other plausible values for the population proportion as well. For example - do you think 75% is a plausible value for the population proportion of all US adults that are right-handed?

 Pro Tip

We just asked another research question! In particular, it’s one that can be addressed using hypothesis testing.

 Question 7

Based on the new research question, “is 75% a plausible value for the population proportion of all US adults that are right-handed?” along with the parameter you defined above, write down the appropriate null and alternative hypotheses. **Hint:** think critically about what we should pick for our alternative hypothesis - there is only one correct answer here!

Solution: As the hint points out, we don’t have a “natural” direction for the alternative - that is, it make sense to use a two-sided alternative:

$$\begin{cases} H_0 : \pi = 0.75 \\ H_A : \pi \neq 0.75 \end{cases}$$

 Caution

I’d like to point out: our original research question, “what proportion of all US adults are right-handed?” is **not** one that can be directly answered using hypothesis testing. Why? Because there are no hypotheses to test!

 Question 8

Use the **One Proportion Applet** to conduct both a simulation-based and theory-based test of the hypotheses you stated in Question 7 above.

- My simulation-based p -value is 0.1671 (answers may vary)
- My theory-based p -value is 0.1307

To show your work, include a screenshot of the applet including any and all inputs/outputs!

Solution:

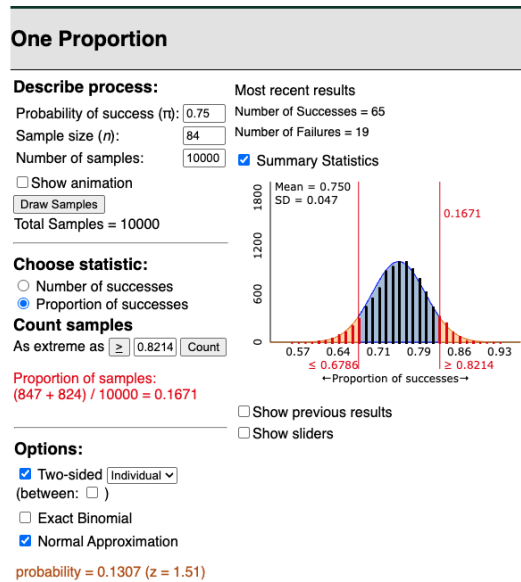


Figure 1: A screenshot from the One Proportion Applet.

Question 9

Based on your results for Question 8 above, is 0.75 a plausible value for the population proportion of all US adults that are right-handed? Explain.

Solution: Both the theory-based and simulation-based p -values constitute weak evidence against the null; therefore, we would fail to reject the null that π is equal to 0.75. Therefore, **yes**; 0.75 is a plausible value for π .

I went ahead and repeated this procedure for a set of different candidate values, and obtained the following results (note that I'm reporting the simulation-based p -values):

Proportion under H_0 :	0.72	0.73	0.74	...	0.89	0.90
p -value	0.036	0.065	0.103	...	0.051	0.027
Reject or Plausible?	Reject	Plausible	Plausible	...	Plausible	Reject

Table 1: Resulting p -values, and whether each specified value is plausible for π or not.

According to Table 1, the 95% confidence interval (using a **significance level** of 0.05) for π is (0.73, 0.89).

Pro Tip

We reject the null whenever the p -value is less than the significance level.

We sometimes call this the **Plausible Values Method**, as it involves checking (using p -values) many different plausible values for the parameter. This method, however, is quite cumbersome - let's work toward developing some shortcuts!

2SD Method

Concept Overview

When a distribution is bell-shaped, as your null distribution should be for this study, approximately 95% of the statistics in the null distribution will fall within two standard deviations of the mean. This implies that 95% of sample proportions will fall within two standard deviations of the population proportion (π), which means that π is within two standard deviations of the observed sample proportion for 95% of all samples.

We can then extend this idea to construct a 95% confidence interval. We can construct a 95% confidence interval of plausible values for a parameter by including all values that fall within 2 standard deviations of the sample statistic. This is sometimes called the **2SD method**. This method is only valid when the null distribution follows a bell-shaped, symmetric distribution. Thus, we can present the 95% confidence interval for the population proportion, π , in symbols as:

$$\begin{aligned} & \hat{p} \pm 2 \cdot (\text{standard deviation of } \hat{p}) \\ &= \hat{p} \pm 2 \cdot SD(\hat{p}) \end{aligned}$$

Recall that:

- $\hat{p}$ is the sample proportion (statistic)
- $SD(\hat{p})$ is the standard deviation of $\hat{p}$
 - One way for us to estimate $SD(\hat{p})$ is to use the value of SD_{NULL} from the simulation
 - Another way is to use a formula (see more information below)
- The mathematical symbol $\pm$ means “plus or minus.” For example, 2 ± 1 means $(2 - 1, 2 + 1) = (1, 3)$

In this formula, $2 \cdot SD(\hat{p})$ is called the **margin of error**. In general, the margin of error is just half the width of the entire interval.

**Question 10**

Calculate the 95% confidence interval for the proportion of all US adults that are right-handed by carrying out the following steps:

- (a) Re-run your simulation using a null value of 0.5 (from your reading of the textbook, you should understand why we're using this value!) What is the value of SD_{NULL} ?

Solution: SD_{NULL} is around 0.055.

- (b) Calculate the margin of error: $2 \cdot SD_{\text{NULL}}$

Solution: $2 \cdot SD_{\text{NULL}} = 2 \cdot (0.055) = \mathbf{0.11}$

- (c) Use this margin of error to produce a 95% confidence interval for π .

Solution:

$$\hat{p} \pm 2 \cdot SD_{\text{NULL}} = (0.8214) \pm (0.11) = \mathbf{(0.7114, 0.9314)}$$

Note: remember to convert your $\hat{p}$ value to a decimal value! (Or, if you want to work with percents, convert the margin of error to percents.)

- (d) Interpret the confidence interval. You are 95% confident that what is between what two values? Make sure your interpretation explicitly references the parameter of interest, along with the population of interest!

Solution: We are 95% confident that the population proportion of all US adults that are right-handed is between 0.7114 and 0.9314 (or, equivalently, between 71.14% and 93.14%)

Though the 2SD method is very useful, it has some limitations:

- 1) What do we do if we don't have a simulated value for SD_{NULL} ?
- 2) What do we do if we want to construct an 90% confidence interval, or a 99% confidence interval (i.e. something other than a 95% confidence interval)?

Let's address the first point. Recall, from Chapter 1, that we know

$$SD(\hat{p}) = \sqrt{\frac{\pi(1 - \pi)}{n}} \quad (1)$$

We used this in the context of hypothesis testing, to conclude that

$$SD_{\text{NULL}} = \sqrt{\frac{\pi_0(1 - \pi_0)}{n}} \quad (2)$$

**Question 11**

- (a) Why can we not, in practice, use the formula in Equation (1) to construct our confidence inter-

val?

Solution: It references the *true* value of π , which is unknown! (Again, as I've said many times now - if we know the true value of π , why would we need to do any hypothesis testing?)

- (b) Why can we not, in practice, use the formula in Equation (2) to construct our confidence interval?

Solution: We do not have a null value, π_0 , anymore, because we are constructing *confidence interval*, not conducting *hypothesis tests*.

So, we instead use the following formula:

$$SD(\hat{p}) \approx \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \quad (3)$$

Okay, so that addresses our first concern above. But what about the second - what do we do if we don't want to be restricted to only constructing 95% confidence intervals?

The answer is: we need to develop a few additional methods to help us with this.

Theory-Based Methods

Recall that our working formula for the confidence interval for π had a multiplier of 2 in front of the SD term. Further recall that the value of 2 came from the fact that, for a bell-shaped distribution, 95% of values fall within 2 standard deviations of the sample statistic. So, to construct, say, a 99% confidence interval, we simply need to figure out: what is the value for which 99% of observations under a bell-shaped distribution fall within that many standard deviations of the sample statistic?

For this class, we'll rely on technology to find the multiplier appropriate for our **confidence level** (the confidence level is just how confident we want to be that our interval contains the true parameter value).



Question 12

Carry out the following steps:

- 1) Navigate to the [Theory-Based Inference Applet](#).
 - 2) Specify the appropriate sample size and sample proportion or sample count, consistent with the data from the Welcome Survey.
 - 3) Press "Calculate."
 - 4) Check the box for "Confidence Interval," and confirm that the default confidence level is 95%. Then press "Calculate CI" to generate a theory-based confidence interval.
- (a) Report the 95% theory-based confidence interval.

Solution: (0.7395, 0.9033)

(b) Is the theory-based confidence interval similar to the one you obtained using the 2SD method?

Solution: Yes, this is quite similar to our 2SD-based confidence interval of (0.727, 0.915).

Concept Overview

The theory-based approach (called a **one-sample z-interval**) is considered valid if there are at least 10 observational units in each category of the categorical variable (i.e., at least 10 successes and at least 10 failures). Because our data has at least 10 successes (69 reported right-handed people) and at least 10 failures (15 reported non-right-handed people), the theory-based approach should produce very similar results to the Plausible Values Method and the 2SD method. In such a case, the theory-based approach is often the most convenient, especially if our confidence level is not 95%.

Concept Overview

In summary, the formula for a theory-based confidence interval for a population proportion is given by

$$\hat{p} \pm \text{multiplier} \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where the multiplier depends on our confidence level.

Pro Tip

- The appropriate multiplier for a 95% confidence interval for a population proportion is around 1.96 (i.e. around 2)
- The appropriate multiplier for a 90% confidence interval for a population proportion is around 1.645
- The appropriate multiplier for a 99% confidence interval for a population proportion is around 2.576



Question 13

(a) Change the confidence level in the applet from 95% to 99% and press the “Calculate CI” button again. Report the 99% confidence interval given by the applet.

Solution: (0.7395, 0.9033)

(b) How does the midpoint of the 99% interval compare to the midpoint of the 95% interval?

☐ The midpoint of the 99% interval is smaller than the midpoint of the 95% interval.

✓ **The midpoint of the 99% interval is the same as the midpoint of the 95% interval.**

- ☐ The midpoint of the 99% interval is the larger than the midpoint of the 95% interval.

(c) How do the the margins of error compare?

- ☐ The margin of error of the 99% interval is smaller than the margin of error of the 95% interval.
- ☐ The margin of error of the 99% interval is the same as the margin of error of the 95% interval.
- ✓ **The midpoint of the 99% interval is the larger than the margin of error of the 95% interval.**

(d) What happens to a confidence interval as the confidence level increases?

- ✓ **The interval gets wider**
- ☐ The interval's width stays the same
- ☐ The interval gets narrower

Interpreting Coverage

I am a stickler about the correct conclusions and interpretations of confidence intervals. The following are **INCORRECT** interpretations of the theory-based 95% confidence interval for π , the proportion of all US adults that are right-handed:

- There is a 95% chance that the population proportion of US adults that are right handed is between 74% and 90%.
 - If you say this, it sounds like there's uncertainty in π , the population proportion - that's not true! π is a fixed number, and it does not vary from sample to sample; therefore, we cannot say that there is a "probability" it lies between two numbers.
- 95% of the sample data falls between 74% and 90%.
 - The interval we constructed is about the *population* proportion of US adults that are right-handed, and gives us no sense of what proportion of sample values fall within the interval.
- We are 95% confident that $\hat{p}$ falls between 74% and 90%.
 - $\hat{p}$ is the **SAMPLE** proportion (of right-handed individuals, in a sample of 84 students) - it's always contained in the confidence interval! Again, we're constructing an interval about the **POPULATION** proportion.

Again, these are all **INCORRECT** interpretations of the confidence interval. Here is the correct one:

We are 95% confident that the population proportion of US adults that are right-handed falls between 74% and 90%.

But, there is still the question of - what do we mean by “95% confident?” Here’s the answer: in the long run, 95% of intervals created with this method succeed in capturing the true value of the parameter. To (hopefully) clarify this a bit, I’ll run through a demo with you.

⚠ Caution

This demo will involve constructing many confidence intervals. It’s important to keep in mind that, in a real-world setting, we only ever get to construct *one* interval, as we often only have *one* sample to work with, and we never really know if our interval contains the true value of the population parameter or not.

If you wish to follow along, I will be making use of the [Simulating Confidence Intervals Applet](#).

Here’s the key takeaway from this would-be demo: If we took repeated random samples from the same population, and each time constructed a 95% confidence interval for π , around 95% of these intervals would cover the true value of π . That is, in the long-run, 95% of intervals constructed this way would cover the true value of the parameter. But, it’s important to note that we ever only get one sample in practice!